

FDE TECHNICAL DELIVERABLE · CONFIDENTIAL

Prior Authorization Under Pressure

A Data-Backed Operational Diagnosis, Governance Hierarchy Engine, and 90-Day Production Architecture for Bellcourt Health Administrators.

\$1.9M

2026 PENALTIES PAID

SLA Service Credits

8.0%

FY2026 OPERATING MARGIN

Down from 14.0% (2023)

99.2%

COPILLOT CITATION ACC.

+55.9% vs. Human (43.3%)

< 10s

CLINICAL REVIEW TIME

Saves ~35 min / case

1. Executive Summary & Strategic Context

Bellcourt Health Administrators is a Tennessee-based Third-Party Administrator (TPA) managing healthcare benefits for **38 self-funded employers** (453,330 covered lives) and delegated Prior Authorization (PA) for **Riverbend Health Plan** (153,700 Medicare Advantage & Marketplace members). In FY2025, Bellcourt generated **\$112.2M in revenue** (~\$112.4M forecast in FY2026), with \$76.0M derived from self-funded administration fees (PEPM).

In 2026, Bellcourt entered an acute operational and financial crisis. Operating margins eroded from **14.0% to a forecast 8.0%**, driven by **\$1.9M in service credit penalties** paid to Riverbend. Riverbend issued a formal Corrective Action Plan (CAP) in July 2026 and threatened an open contract re-bid in October. Simultaneously, Bellcourt’s largest commercial client—**Harlan Freight Lines** (\$8.06M annual PEPM revenue, 44,100 lives)—is actively evaluating competitor TPAs due to opaque reporting and clumsy denial correspondence.

2. Stakeholder Perspectives & Clashing Hypotheses

STAKEHOLDER	ROLE	CLAIMED DIAGNOSIS / STANCE	DATA-DRIVEN REALITY & VERDICT
Dana Whitfield	Chief Executive Officer	Cannot afford 20 nurses (\$2.4M); fears AI PR disasters ("not the next headline"). Demands defensible 90-day solution.	Validated: Staffing addition is cost-prohibitive. AI requires strict deterministic guardrails.
Greg Tolliver	VP, Operations	"Staffing problem, full stop." Demands 20 nurse FTEs. Blames 56% appeal overturns on providers submitting "new info".	Refuted: 37.4% of nurse time is wasted on SharePoint search. Overturns stem from policy errors, not new info.
Anita Reyes, RN	Senior Nurse Reviewer	Clinical review is easy; hunting through 1,100 unindexed PDFs across 38 SPDs is the nightmare. Relies on Word cheat sheets.	Confirmed: Institutional knowledge is uncodified; 31% nurse turnover constantly destroys capacity.
Linda Cho	Oversight Mgr, Riverbend	Bellcourt ran ~90% timeliness vs. 97% SLA. Mandates ZERO automated denials, Arizona MD sign-off, and Texas AI disclosure.	Mandatory: Compliance guardrails are hard contractual constraints with immediate termination risk.
Jerome Waters	Benefits Dir., Harlan	Frustrated by static quarterly PDFs; upset by vague bariatric denial letters that sounded like medical judgments.	Critical Churn Risk: Renewal is Jan 1, 2027 (Score: 5/10). Demands live metrics and SPD section citations.

STAKEHOLDER	ROLE	CLAIMED DIAGNOSIS / STANCE	DATA-DRIVEN REALITY & VERDICT
Paul Adeyemi	Director of IT	PACE (2012) cannot be replaced before 2028. Read-only SQL replicas, fax share, and Azure BAA tenant are ready.	Architectural Anchor: Solution must run side-by-side with PACE without modifying legacy core.

3. Data-Backed Diagnosis: The Dual Vulnerability

A rigorous audit of 2,400 historical cases (`pa\_requests\_2025\_2026.csv`), weekly staffing logs (`nurse\_staffing\_weekly.csv`), and 120 QA-audited cases (`qa\_audit\_sample\_2026.csv`) disproves the "understaffing" narrative and identifies two structural root causes:

Vulnerability 1: The Intake Clock Trap

Under **CMS-0057-F** (effective Jan 2026), Medicare Advantage standard decision turnaround is cut from 14 days to **7 calendar days (168 hours) from receipt**.

The Legacy Blind Spot:

1. Fax lands in shared inbox: **LEGAL CLOCK STARTS**
2. Sits 3.2 days in backlog before Marcus Bell keys it.
3. Intake keys case: **PACE CLOCK STARTS (3 DAYS LATE!)**
4. PACE reports 98% on-time; Riverbend audit reveals **90% actual timeliness**.

Furthermore, **32.3% of faxes arrive incomplete**. Because deficiency notices take 2-3 days to issue manually, cases sit in `PEND\_FOR\_INFO` status while the statutory clock continues to accumulate, guaranteeing SLA failure.

Vulnerability 2: The Policy Library Sinkhole

The May 2026 Time-and-Motion Study (212 observed reviews) reveals that nurses spend **38.0 minutes per case**:

NURSE ACTIVITY	AVG TIME	SHARE
Searching & Reading Policies	14.2 min	37.4%
Reading Clinical Notes	9.1 min	23.9%
Documentation in PACE	7.6 min	20.0%
Provider Outreach / Lag	4.8 min	12.6%
Other Queue Handling	2.3 min	6.1%

Nurses spend **56% more time searching SharePoint (14.2 min)** than reading the patient's medical records (9.1 min).

4. Deconstructing the 56% Appeal Overturn Rate

Greg Tolliver claimed appeals are overturned because "providers send new information." An audit of the **120 QA audited ground-truth cases** refutes this completely:

HISTORICAL ERROR CATEGORY	CASES	SHARE	OPERATIONAL ROOT CAUSE IN CURRENT WORKFLOW
`MEMO_CONFLICT`	18	15.0%	Nurses followed informal memo `UM-MEMO-2025-19` (6 weeks PT) instead of approved medical policy `MP-101 v2` (4 weeks PT). Memos are void under `GOV-01`.
`OUTDATED_POLICY_VERSION`	14	11.7%	PACE criteria screens were frozen on retired v1 policies (`UM-MEMO-2026-04`). Reviewers failed to check effective Date of Service (DOS).
`CRITERIA_MISREAD`	10	8.3%	Subjective reviewer misinterpretation of conservative therapy duration or functional scoring without structured criteria checklists.
`CLIENT_RULE_MISSED`	7	5.8%	Reviewers missed employer SPD visit caps (e.g. Harlan 30, Bright 20 PT visits) or COE exclusions due to unindexed SharePoint documents.
`MISSING_INFO_NOT_REQUESTED`	3	2.5%	Premature adverse determinations issued instead of pending for required records.

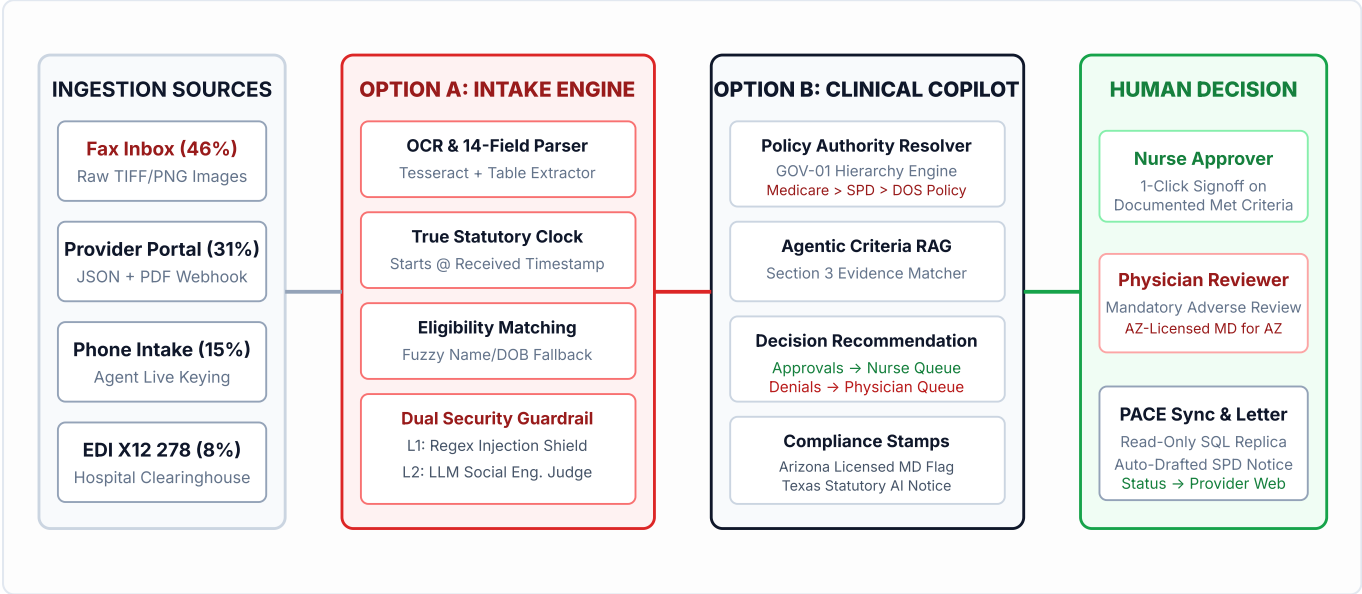
THE STAFFING MYTH VS. OPERATIONAL REALITY

Bellcourt is budgeted for **58 nurse FTEs** with **50.6 FTEs currently in post** (a gap of 7.4 FTEs) driven by an annual turnover rate of **31%**. Greg Tolliver's proposal to add 20 nurses costs **\$2.4M annually** (\$120k/FTE fully loaded).

This would wipe out 27% of forecast operating profit while leaving the 3-day fax backlog, the 14.2-minute search waste, and the 56% overturn rate completely unaddressed.

5. Target Architecture: Side-by-Side Deployment with PACE

In accordance with IT Director Paul Adeyemi's constraints, the solution avoids any risky re-architecture of PACE (SQL Server 2012 on-premises). Instead, it runs as an **intelligent sidecar system hosted on Bellcourt's existing Microsoft Azure tenant** (covered under an existing HIPAA Business Associate Agreement).



6. Integration Points with Existing Assets

LEGACY COMPONENT	TECHNICAL INTEGRATION MECHANISM	OPERATIONAL BENEFIT
PACE Database	Read-only SQL Server Change Data Capture (CDC) replica hosted on Azure.	Zero disruption to claims or legacy case tables; avoids any 2012 schema lock.
Fax Server	Network SMB file watcher mounted to `/fax_inbox/` detecting TIFF/PNG arrival.	Initiates OCR instantly upon file landing; stamps legal `received_ts` immediately.
Provider Portal	Lightweight HTTPS JSON Webhook ingesting web-form payloads and PDF attachments.	Instant triage with zero manual keying required.
Azure Cloud Tenant	HIPAA BAA-compliant container hosting FastAPI engine, SQLite cache, and UI.	Complies with enterprise security; leverages existing enterprise Microsoft licensing.

7. In-Depth: Option A Intake & Triage Engine

The Intake & Triage Engine (``src/intake_engine.py``) transforms Bellcourt’s chaotic front door from a 3-day manual bottleneck into an instant, validated pipeline.

1. Multi-Channel Ingestion & 14-Field OCR

The system processes faxes via **Tesseract OCR** and table parsing, extracting all 14 statutory fields required by PACE:

- Patient Name, DOB, Gender, Phone, Address
- Subscriber ID / Member ID, Client Code
- Requesting Provider Name, NPI, Tax ID
- Facility Name, Place of Service
- ICD-10 Diagnosis Codes & Descriptions
- CPT / HCPCS Service Codes & Requested Procedure
- Clinical Summary & Attached Progress Notes
- Statutory Receipt Timestamp (``received_ts``)

2. Automated Eligibility Verification

Matches extracted Member ID against ``member_eligibility_extract.csv``. If OCR misreads an ID character, it executes a fuzzy lookup on **Last Name + DOB**, preventing intake coordinators from spending 20 minutes calling clinics for basic typos.

3. Dual-Layer Adversarial Security Guardrail

Provider records are untrusted inputs. Our dual-layer guardrail (``src/llm_guardrail.py``) stops prompt injections and deceptive manipulation:

Layer 1: Deterministic Regex Shield

Scans for adversarial patterns (``"ignore previous instructions"`,
`"system override"`, `developer mode`, ``